long-range missiles, and advanced air-defense weapons. Its modular design ensures compatibility with future AI modules, next-generation sensors, and autonomous operational frameworks.

Strategically, the KF-21 program marks South Korea's emergence as a next-generation defense innovator, reducing reliance on foreign suppliers and boosting its aerospace export potential. It also signifies a technological partnership within Asia, as Indonesia contributes financially and industrially to the program. With continuous upgrades planned through the Block III phase, the KF-21 is poised to evolve into a fully fifth-generation fighter, bridging the technological divide between fourth-generation legacy fighters and the autonomous air systems of the future.

Figure 6.3 illustrates the KF-21 Boramae, South Korea's advanced multirole stealth-capable fighter aircraft developed by Korea Aerospace Industries (KAI). The labeled diagram highlights its core systems, including the internal weapon bays, avionics suite, internal IRST (Infrared Search and Track) sensor, and planned Block III upgrades. The figure emphasizes the aircraft's stealth-optimized design, characterized by angled fuselage geometry, composite airframe materials, and concealed weapon compartments for reduced radar cross-section.

